demonstrated

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SOURCE: MEPC.373(80).pdf
3.15 The device should demonstrate ongoing compliance with the emission limits to air at all times when in operation, including warm-up and shutdown phases
Means by which ongoing compliance with the emission limits is to be demonstrated
3.16 The means by which ongoing compliance with the emissions to air are monitored should produce reliable measurement data
Means by which monitoring equipment, and any associated equipment, are operated, zero and span checked, maintained and serviced to achieve the required measurement performance
3.17 The device should be shut down in a manner which ensures that all thermal processes are terminated and that the device and associated exhaust system to atmosphere is purged of all residual gases
Operating procedures and procedures relating to the shutdown of the device
3.18 The means and procedures for the removal of solid residues from the device should ensure that these are fully and securely contained for landing ashore
Operating means and procedures in respect of the removal of solid residue material from the device and subsequent onboard storage prior to discharge ashore

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SOURCE: MEPC.373(80).pdf
should ensure that these are fully and securely contained for landing ashore
Operating means and procedures in respect of the removal of solid residue material from the device and subsequent onboard storage prior to discharge ashore
3.19 The discharge water arrangements should be prepared, operated and shutdown such that the emission to sea limits are met under all operating conditions including

during the device warm -up and shutdown phases
Operating procedures of the discharge water arrangements, including control and monitoring functions, relating to the preparation for use, in -service application and shutdown
3.20 Discharge water PAH limit should be expressed as a concentration (µg/litre) as appropriate to the device across its operating range
PAH limit(s) should be given against discharge water flow rate(s)
3.21 The means by which ongoing compliance with the discharges to sea is monitored should produce reliable measurement data
Means by which monitoring equipment, and any associated equipment, are operated, zero and span checked maintained and serviced to achieve the required measurement performance

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SOURCE: MEPC.373(80).pdf
should produce reliable measurement data
Means by which monitoring equipment, and any associated equipment, are operated, zero and span checked maintained and serviced to achieve the required measurement performance
3.22 Additional Functional Objectives related to device operations, as applicable to this type of device and relevant in terms of meeting the emission limits and the objectives of regulation 4.4 of MARPOL Annex VI, should be identified
Resolution of those additional Functional Objectives
4. Record-keeping

Functional objective Content of TWTD Technical Report

4.1 There should be an Events Record for

each TWTD device installed. That Events Record is to cover all phases of operation of the device when in service
What the Events Record is to include and the manner of its recording
4.2 There should be retained records in respect of emissions to air
Form of records which are required to be kept demonstrating the performance and self -checking

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SOURCE: MEPC.373(80).pdf
What the Events Record is to include and the manner of its recording
4.2 There should be retained records in respect of emissions to air
Form of records which are required to be kept demonstrating the performance and self -checking functions against respective limits showing that the device performed as required set against the recorded Event Record points
4.3 There should be records of the oxygen content, temperature and pressure values, as applicable to the principle of operation, through the device showing that it operated within the required profiles
Form and extent of records which should be kept demonstrating that the required oxygen content, temperature and pressure values were achieved set against the recorded Event Record points
4.4 There should be records in respect of emissions to sea
Form of records which are required to be kept demonstrating the performance and self -checking functions against respective limits showing that the device performed as required set against the recorded Event Record points
4.5 There should be records of solid, and any other, residue materials or related liquids discharged ashore

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SOURCE: MEPC.373(80).pdf
functions against respective limits showing that the device performed as

required set against the recorded Event Record points

4.5 There should be records of solid, and any other, residue materials or related liquids discharged ashore

Related record-keeping requirements

4.6 Records should be against date and UTC. These records should be retained on board at least 18 months from date of recording. If the recording device is changed over that period, it should be ensured that the required data is retained on board and available as required. The recording device should be capable of producing reports as required demonstrating past performance

Means by which the required records are to be recorded and retained on board in a tamper-proof manner.

The extent and form of the reports that the recording device is capable of producing